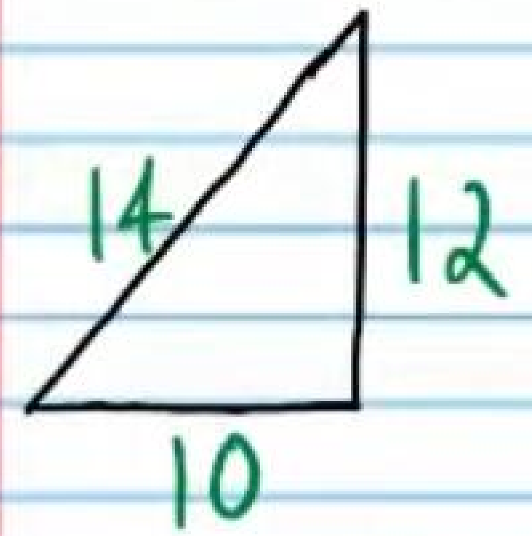
Name
Date
Course

Practice Geometry Test IV (Version 2)

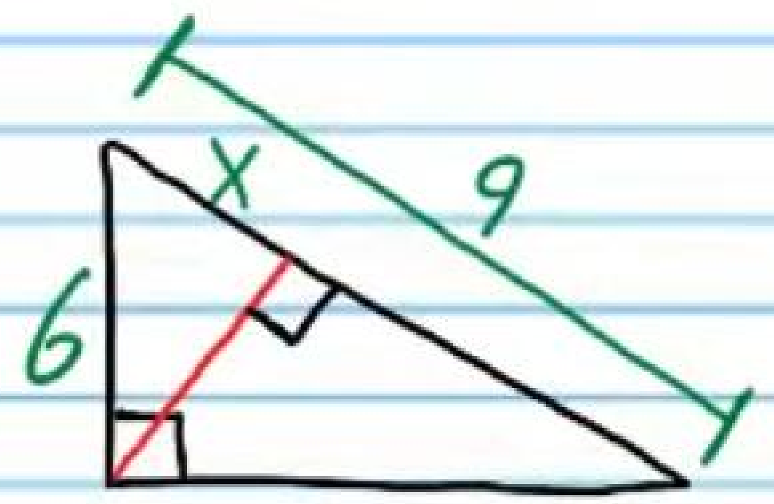
Instructions

I) Write your name, the date, and the course title (i.e. Geometry).
II) Show work when necessary.
III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) Each problem is worth four points for a total of 100 points.
V) A calculator will be necessary for some problems on this test.

① Is the following polygon an acute, obtuse, or right triangle?

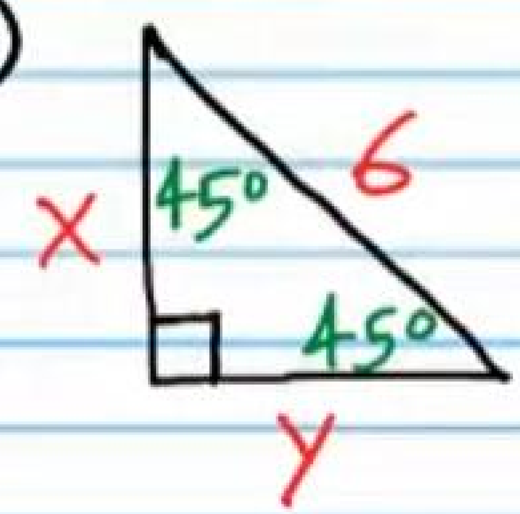


② Find x below.



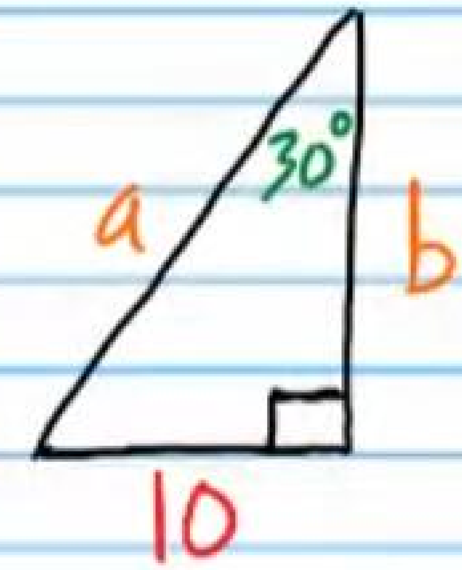
Find the unknown values.

③



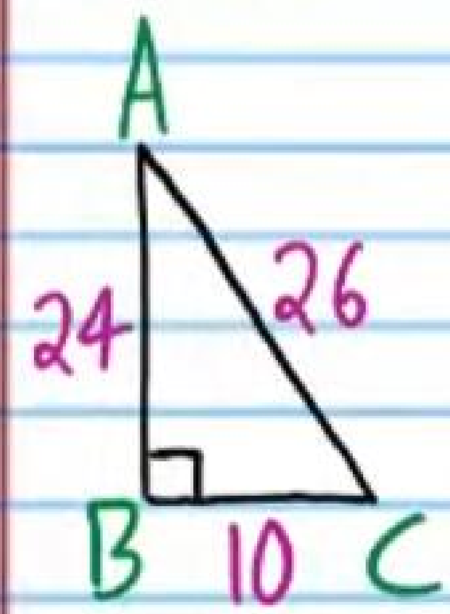
i) x ii) y

④

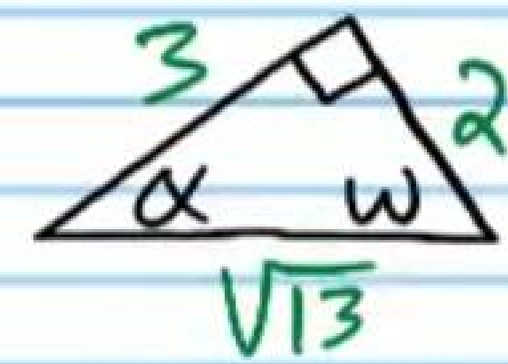


i) a ii) b

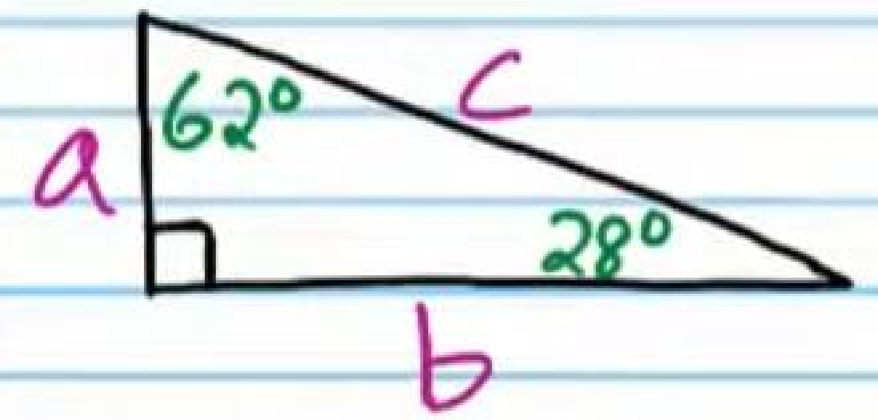
⑤ i) For the triangle below, find cos A.



ii) For the triangle below, find sin α.

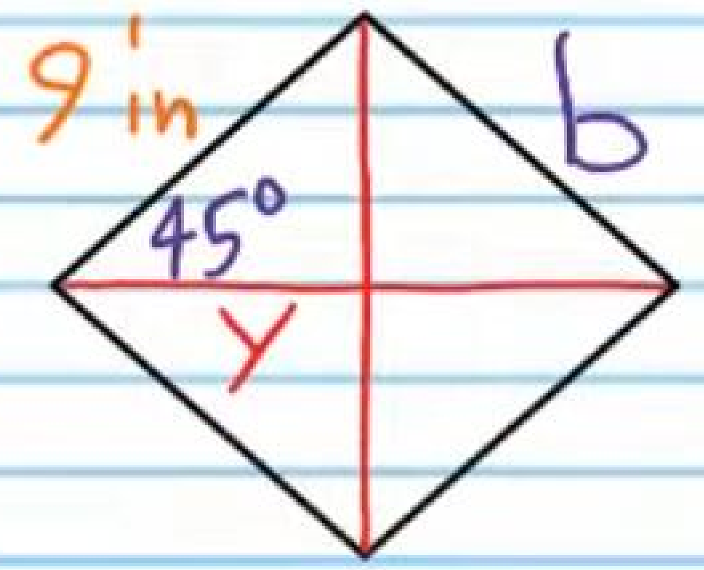


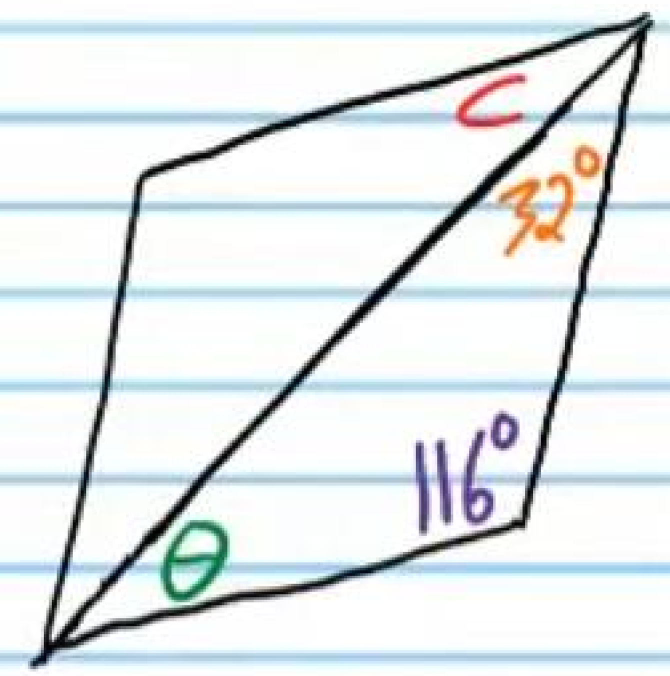
⑥ i) For the triangle below, find tan 62°. Your answer should have letters. Do not use a calculator.



ii) Use a calculator to find tan 65°. Round to the nearest thousandth.

⑦ Find the unknowns in each rhombus below.





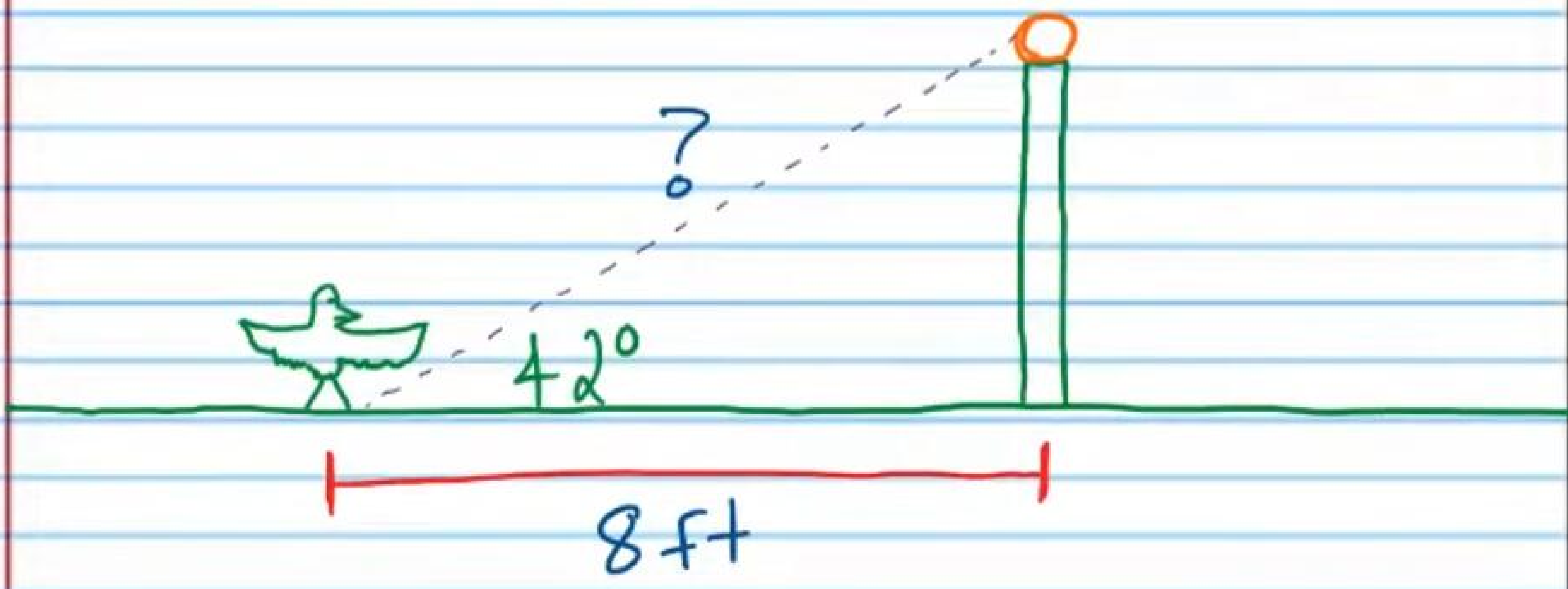
i) y

ii) b

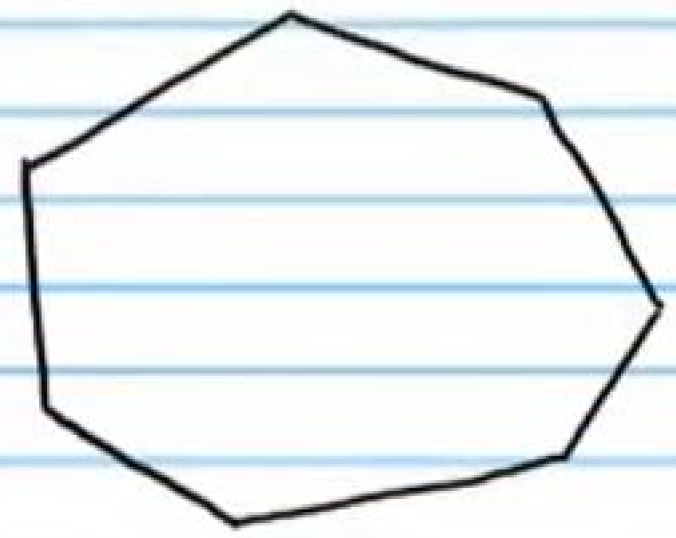
iii) c

iv) θ

⑧ A bird is standing on the ground eight feet from a light pole. The angle of elevation from the bird to the top of the pole is 42°. What is the shortest distance that the bird will have to fly to reach the top of the pole?



⑨ Find the sum of the measures of the interior angles of the following polygon.

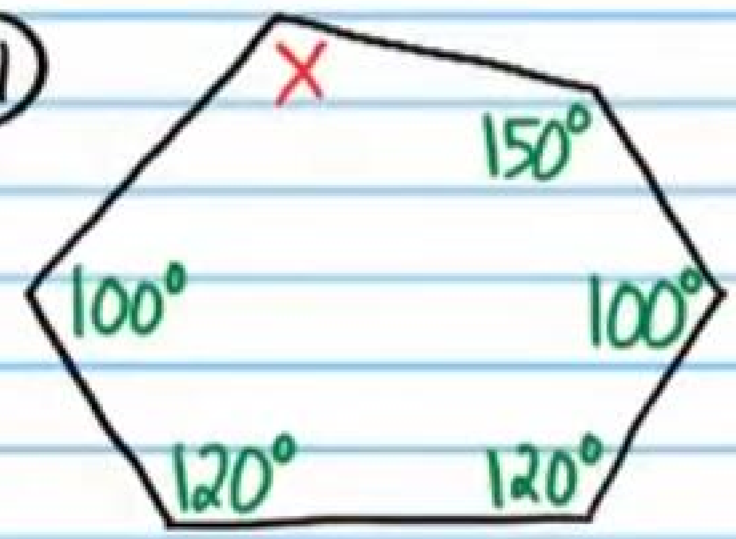


⑩ If the number below is the sum of the measures of the interior angles of a convex polygon, find the number of sides of the polygon.

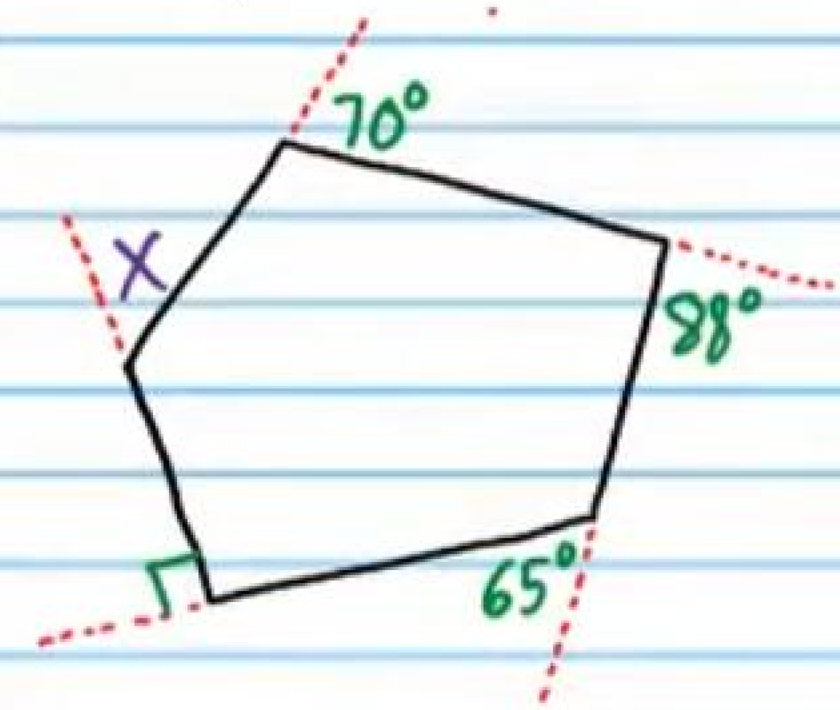
2340°

Find the unknowns in each problem below.

⑪



⑫



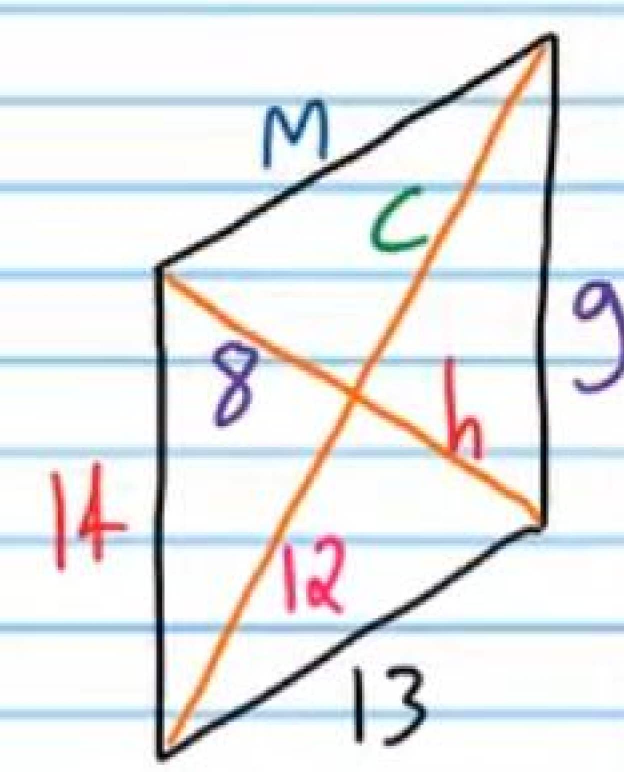
⑬ Find the unknown values of the parallelogram below.

i) h

ii) c

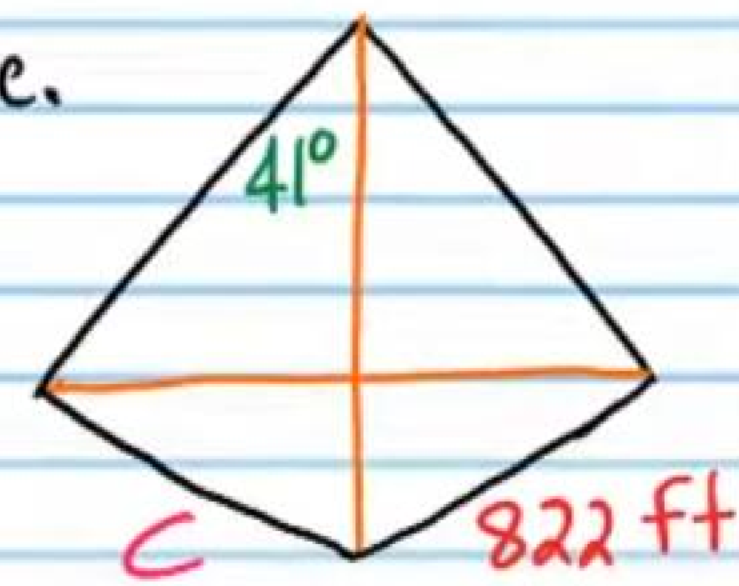
iii) g

iV) m



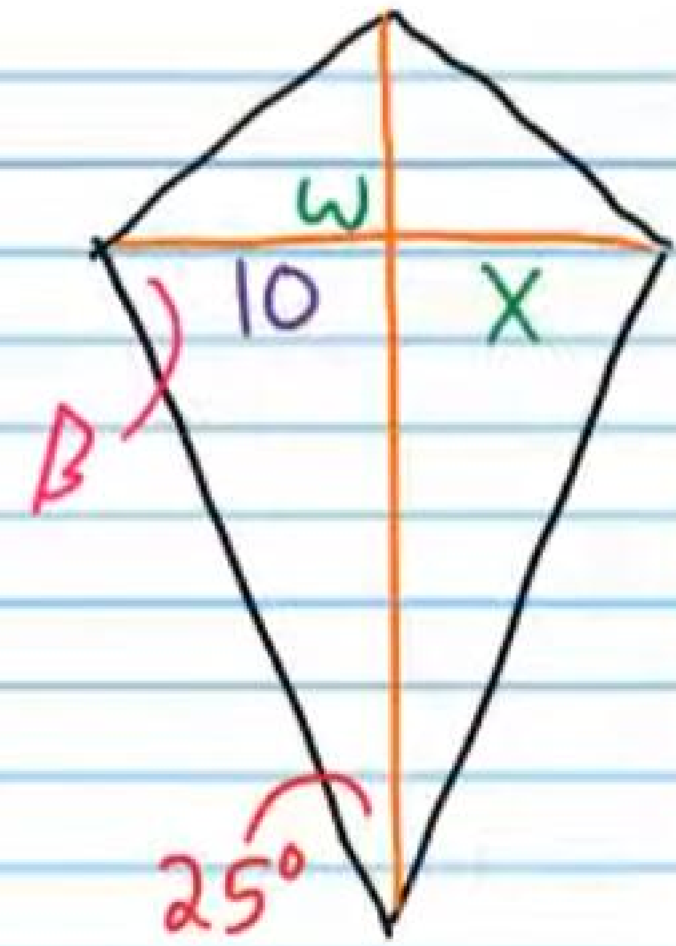
⑭ Find the unknowns in each kite.

i) c



ii) x iii) w

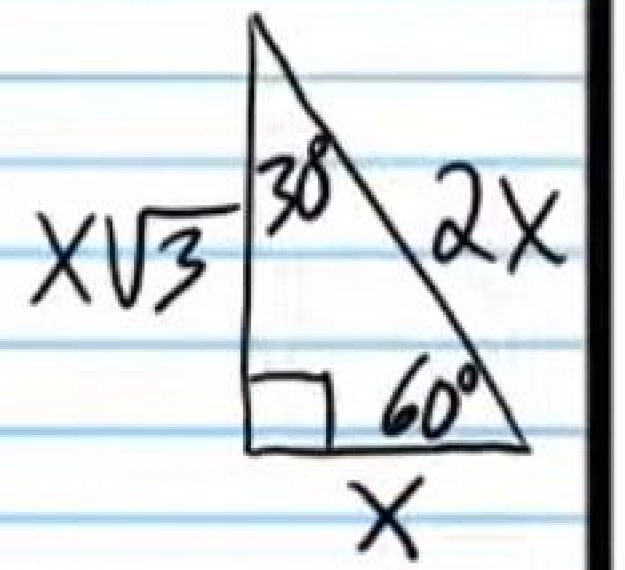
iv) β



(15) Given the triangle below, find the following values without a calculator.

i) $\tan 30°$

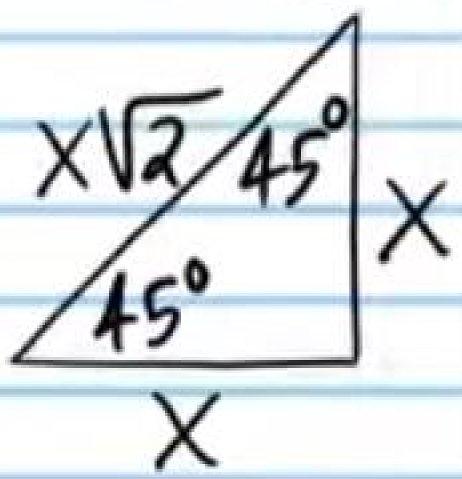
ii) $\cos 60°$



Given the triangle below, find the following values without a calculator.

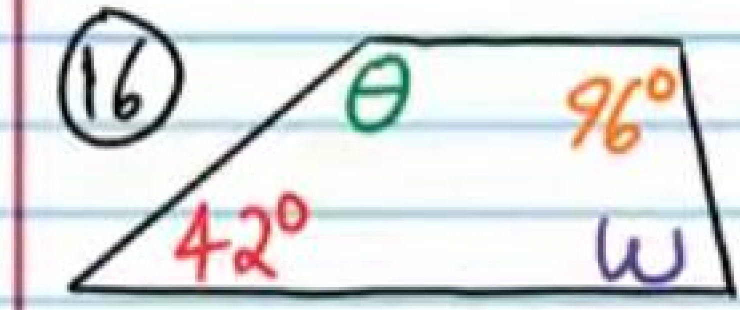
iii) $\tan 45°$

iv) $\cos 45°$



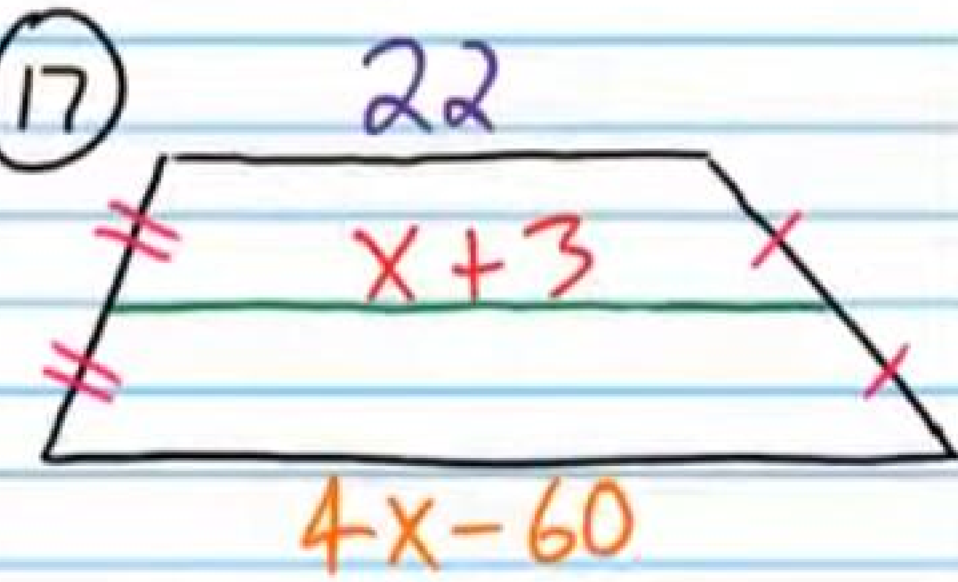
Find the unknowns in each trapezoid below.

(16)

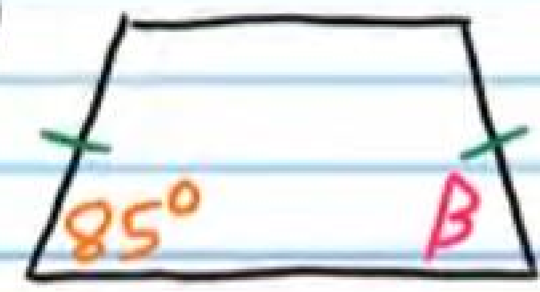


i) θ ii) w

(17)



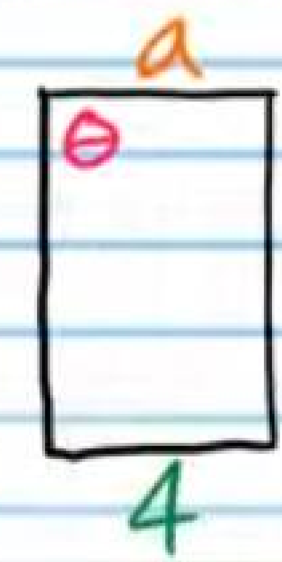
(18)



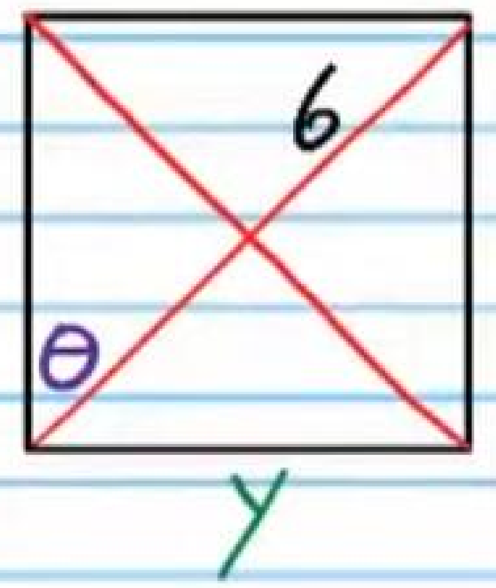
Find the unknowns in the rectangle below.

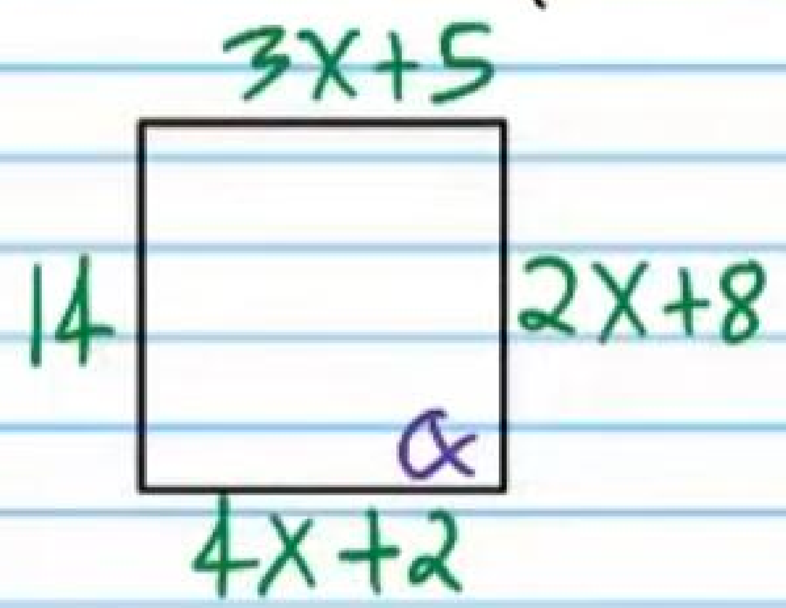
(19) i) a

ii) θ



(20) Find the unknowns in each square below.





i) y

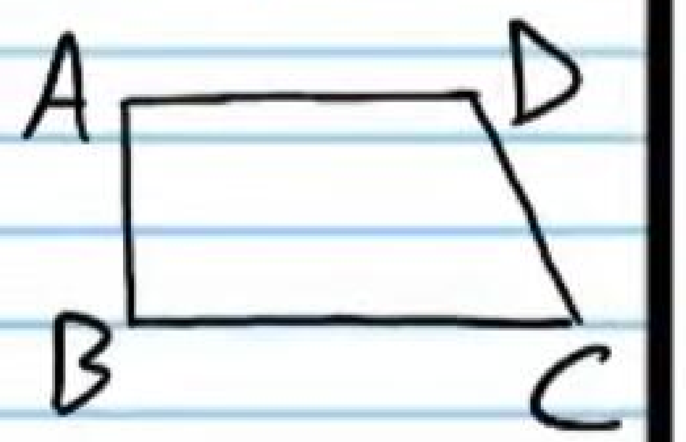
ii) θ

iii) α

iv) x

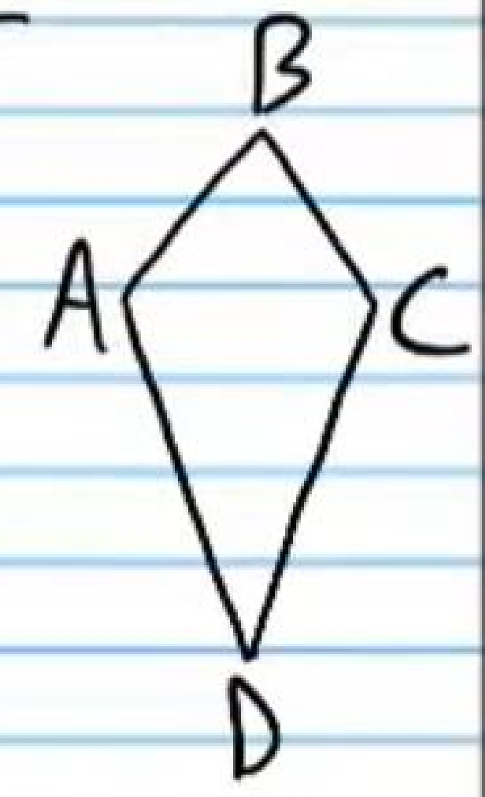
Show that the following statements are true using two-column proofs. You might not use the same number of steps written in Roman numerals. You may use more or less.

(21) If the quadrilateral below is a trapezoid, then $\overline{AD} \parallel \overline{BC}$. Yes. There are only two steps.



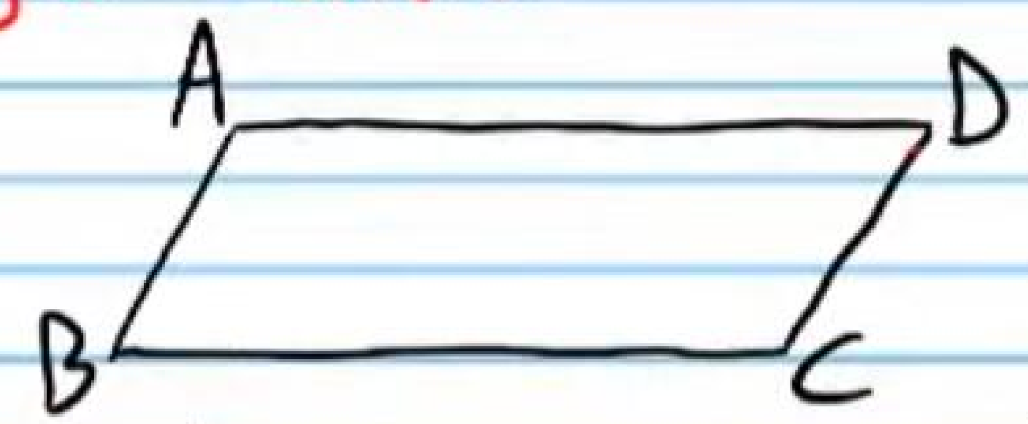
Statement	Reason
I)	
II)	

(22) If the quadrilateral below is a kite, then $\triangle ABD \cong \triangle CBD$.



Statement	Reason
I)	
II)	
III)	
IV)	

(23) If the quadrilateral below is a parallelogram, then $\angle A$ & $\angle D$ are supplementary, $\angle B$ & $\angle C$ are supplementary, $\angle A$ & $\angle B$ are supplementary, and $\angle D$ & $\angle C$ are supplementary. This is the Parallelogram Consecutive Angles Theorem.

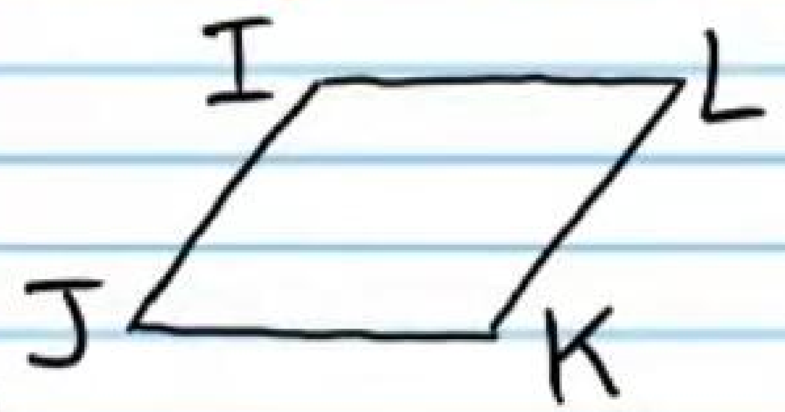


Statement	Reason
I)	
II)	
III)	

IV)

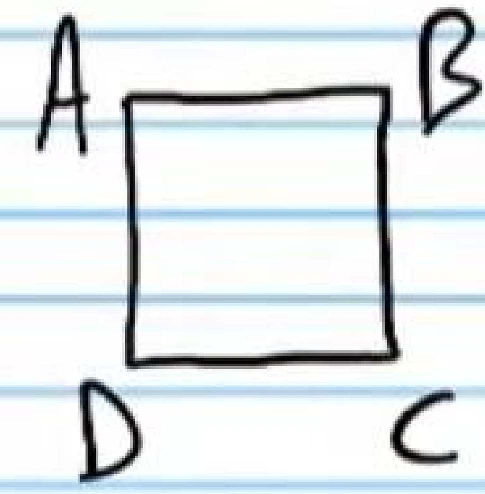
V)

(24) If the quadrilateral IJKL below is a rhombus, then its diagonals bisect the inner angles (i.e. $\angle JIK \cong \angle LIK$, $\angle JKI \cong \angle LKI$, $\angle ILJ \cong \angle KLJ$, and $\angle IJL \cong \angle KJL$).
This is the Rhombus Angle Bisector Theorem.



Statement	Reason
I)	
II)	
III)	
IV)	
V)	

(25) If the quadrilateral below is a square, then it is a parallelogram. Hint: You must use the Consecutive Interior Angles Converse Theorem.



Statement	Reason
I)	
II	
III)	
IV)	
V)	
VI)	
VII)	
VIII)	

Practice Geometry Test IV (Version 2) Answers

① $\boxed{\text{Acute}}$

② $\boxed{x=4}$

③ i) $\boxed{x = 3\sqrt{2}}$

ii) $\boxed{y = 3\sqrt{2}}$

④ i) $\boxed{a = 20}$

ii) $\boxed{b = 10\sqrt{3}}$

⑤ i) $\boxed{\frac{12}{13}}$

ii) $\boxed{\frac{2\sqrt{13}}{13}}$

⑥ i) $\boxed{\frac{b}{a}}$

ii) $\boxed{2.145}$

⑦ i) $\boxed{y = 9\sqrt{2}/2 \text{ in}}$

ii) $\boxed{b = 9 \text{ in}}$

iii) $\boxed{C = 32^\circ}$

iv) $\boxed{\theta = 32^\circ}$

⑧ $\boxed{10.765 \text{ ft}}$ * The problem does not ask to round to a certain decimal place, so 10.8 or 10.77 are also acceptable.

⑨ $\boxed{900^\circ}$

⑩ $\boxed{15}$

⑪ $\boxed{x = 130^\circ}$

⑫ $\boxed{x = 47^\circ}$

⑬ i) $\boxed{h = 8}$

ii) $\boxed{c = 12}$

iii) $\boxed{g = 14}$

iv) $\boxed{m = 13}$

⑭ i) $\boxed{c = 822 \text{ ft}}$

ii) $\boxed{x = 10}$

iii) $\boxed{w = 90^\circ}$

iv) $\boxed{\beta = 65^\circ}$

⑮ i) $\frac{1}{\sqrt{3}} = \boxed{\frac{\sqrt{3}}{3}}$

ii) $\boxed{\frac{1}{2}}$

iii) $\boxed{1}$

iv) $\frac{1}{\sqrt{2}} = \boxed{\frac{\sqrt{2}}{2}}$

⑯ i) $\boxed{\theta = 138^\circ}$

ii) $\boxed{w = 84^\circ}$

⑰ $\boxed{x = 22}$

⑱ $\boxed{\beta = 85^\circ}$

⑲ i) $\boxed{a = 4}$

ii) $\boxed{\theta = 90^\circ}$

⑳ i) $\boxed{y = 6\sqrt{2}}$

ii) $\boxed{\theta = 45^\circ}$

iii) $\boxed{\alpha = 90^\circ}$

iv) $\boxed{x = 3}$

㉑

Statement	Reason
I) Quadrilateral ABCD is a trapezoid.	Given
II) $\overline{AD} \parallel \overline{BC}$	Def. of a ⏢.

(22)

Statement	Reason
I) Quadrilateral ABCD is a kite.	Given.
II) $\overline{AB} \cong \overline{BC}$ & $\overline{AD} \cong \overline{CD}$.	Def. of a kite.
III) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$.
IV) $\triangle ABD \cong \triangle CBD$	SSS

(23)

Statement	Reason
I) ABCD is a parallelogram.	Given.
II) $\overline{AB} \parallel \overline{DC}$	Def. of a ▱.
III) $\angle A$ & $\angle D$ are supplementary and $\angle B$ & $\angle C$ are supplementary.	CIAT
IV) $\overline{AD} \parallel \overline{BC}$	Def. of ▱.
V) $\angle A$ & $\angle B$ are supplementary and $\angle D$ & $\angle C$ are supplementary.	CIAT

(24)

Statement	Reason
I) Quadrilateral IJKL is a rhombus.	Given.
II) $\overline{JI} \cong \overline{IL} \cong \overline{JK} \cong \overline{KL}$.	Def. of a rhombus.
III) $\overline{JL} \cong \overline{JL}$ & $\overline{IK} \cong \overline{IK}$	Ref. Prop of $\cong$.
IV) $\triangle JIL \cong \triangle JKL$ & $\triangle JIK \cong \triangle LIK$	SSS
V) $\angle IJL \cong \angle KJL$, $\angle ILJ \cong \angle KLJ$, $\angle JIK \cong \angle LIK$, and $\angle JKI \cong \angle LKI$	CPCTC

(25)

*You may choose any three interior angles.

Statement	Reason
I) Quadrilateral ABCD is a square.	Given.
II $\angle A$, $\angle B$, & $\angle C$ are right angles.	Def. of a □.
III) $m\angle A = 90°$, $m\angle B = 90°$, and $m\angle C = 90°$	Def. of rt $\angle$.
IV) $m\angle A + m\angle B = 90 + 90$ and $m\angle B + m\angle C = 90 + 90$.	Subs. Prop of =.
V) $m\angle A + m\angle B = 180$ and $m\angle B + m\angle C = 180$.	Simplification.
VI) $\angle A$ & $\angle B$ are supplementary and $\angle B$ & $\angle C$ are supplementary.	Def. of supp. $\angle$'s.
VII) $\overline{AD} \parallel \overline{BC}$ & $\overline{AB} \parallel \overline{DC}$	CIACT
VIII) Quadrilateral ABCD is a parallelogram.	Def. of a ▱